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Original Article

Imparting child nutritional assessment skills among medical students using hybrid problem-based learning approach in a community setting

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ABSTRACT

Background: Malnutrition affects up to a third of children in India, with severe and acute malnutrition prevalent among under five children. Nutritional assessment skills for detecting malnutrition in children in primary care settings are vital. Hybrid problem-based learning (HPBL) is an innovative, collaborative, and adaptable instructional learning strategy that can be used to teach medical students clinical skills in a community setting.
Methods: A two-month quasi-experimental study was undertaken in a rural setting with third-year medical students. Faculty members were sensitized and subject experts developed a training module addressing the knowledge, attitude, communication, and practice domains. The students underwent a 3-week training module where pre-testing, case presentation, and group formation in first week, an anchoring lecture, tutorial and self-directed learning and role-play by students in subsequent week, and in last week, case discussion, post-testing, and feedback rounds were done.
Results: In all domains, knowledge (3.8, 0.01), practice (4.3, 0.01), attitude and communication (3.7, 0.01), and proportional satisfactory responses, the HPBL approach resulted in a significant improvement in nutritional assessment competency. Teachers preferred the practical and engaging character of the approach, stating that doubts and questions were better addressed and that they would use it to teach similar topics.
Conclusion: HPBL is an excellent teaching method for clinical skills, like nutritional assessment in simulated/field settings. The novel teaching-learning approach was well received by students and faculty members. Learning outcomes and satisfaction rates enhanced in students and faculty were encouraged to apply the approach to other topics.
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Introduction

Child growth and health are recognized as important public health indicators for monitoring nutritional status and health in a population. India remains home to nearly one-third of the world's children afflicted with severe and acute malnutrition (SAM). According to the NFHS-5 fact sheet, the prevalence of underweight amongst under five children for the years 2019–2021 is 32.1%. Similarly 35.5% children were reported as stunted (height-for-age) and 19.3% wasted (weight-for-height).[1]

Nutritional assessment is a very important screening and diagnostic tool for the detection of malnutrition among children. A comprehensive nutritional assessment should include mnemonic "ABCD" methods which include anthropometric examination, biochemical/laboratory testing, clinical examination, and dietary assessment.[2] However, currently employed teaching methods for nutritional assessment are largely didactic and lack a skill-based approach and practical orientation to the learners. This, in turn, cascades to the incompetency of Indian medical graduates in effectively performing their role of first contact physician in managing such cases in primary care settings. Unmet training needs lead to a lack of motivation and confidence in providing skilled care to malnourished children which translates into poor rapport with community and health workers and high malnutrition attributed to morbidity and mortality in children.

Hybrid problem-based learning (HPBL) is a class-based teaching method. During the HPBL method of teaching, the problems are solved in groups, whereas traditionally, lectures are used to clarify difficult topics and share fundamental concepts.[3]

A hybrid problem-oriented approach is one of the most effective ways for students to explore the various facets of "real-life issues" confronted by them in their careers. In addition to problem-solving skills, case discussions promote collaborative learning, teamwork, reflection, and self-directed learning. Such approaches have been used in past for teaching nutrition and its applied aspects to medical students with significant learning outcomes. Such innovative teaching-learning methods have been recommended for integration into the current medical curriculum as they will enable skilled management of malnutrition in primary care settings and early identification and referral of high-risk children thereby reducing malnutrition attributed burden.

Previous studies have suggested positive feedback from students and significant improvement in students' knowledge.[4] It was also reported that the HPBL catalyzes learning among interns with minimum efforts by facilitators or teachers. Interns can be exposed to a hands-on learning experience in population-based research, on a systematic basis, without additional resources.[5] The positive effects of HPBL in terms of additional learning resources, critical thinking, and fun learning experiences reported by students emphasize the effectiveness of the technique.[6] The current study was designed to train phase III medical students for child nutritional assessment using a hybrid PBL approach in the community setting and assess learning outcomes in terms of improved cognitive, psychomotor, and affective performance.

Material and methods

Study design

Quasi-experimental study (Pre-post design).

Study duration

2 months (August 2019–September 2019).

Study sample size and sampling

A batch of phase III medical students (n = 25) posted in the Department of Community Medicine was included in the study by complete enumeration after their consent.

Study setting

The study was conducted in the field practice area of the Rural Health Training Center affiliated with the Department of Community Medicine of the institution.

Faculty sensitization was done in the Department of Community Medicine. Followed by that, a total of 5 subject experts developed a comprehensive competency-based training module, which covered knowledge, attitude, communication, and practice domains related to nutritional assessment. This module was used for training the study subjects.

The pre- and post-test questionnaire, feedback questionnaire for students and faculty, and observation checklist were developed. The questionnaire and checklist were pilots tested for internal reliability (Cronbach's alpha = 0.85) and external validity through expert review. The pre and post questionnaire contained items to assess knowledge and attitude and the checklist contains score items to assess the psychomotor and communication skills of trainee students. For the assessment of knowledge, the questions based on the growth chart, SAM assessment, nutritional and dietary assessment, and clinical signs symptoms were framed. The practical approach was assessed by medical history, physical examination including anthropometry, clinical diagnosis, and management. The observation checklist contains a rating scale adapted from the Kalamazoo communication checklist.[7] Observation checklists were developed for comprehensive skill assessment of the knowledge, attitude, communication, and practice of students. The Knowledge of students was assessed using 5 questions (9 marks), practical using 5 questions (25 marks), and attitude and communication using 7 questions (35 marks). The observation checklist had items with a maximum score of 1 and 5; satisfactory scores were adjudged either 1 out of 1 or 3 out of 5 on individual items in the three learners' domains. The feedback questionnaire was having items of satisfaction regarding the HPBL session.

The module was used as an intervention method for the students. The HPBL model (Fig. 1) focused on self-directed and collaborative learning was adopted for training.[8]

The study subjects underwent a 3-week training program using the module. During 1st week (session 1), pretest assessment, introduction to the nutritional assessment case, and group formation were done. In the 2nd week (session 2), intervention in the form of an interactive tutorial, learning activities like self-direct learning, anchoring lecture on nutritional assessment, and role-play by students for the

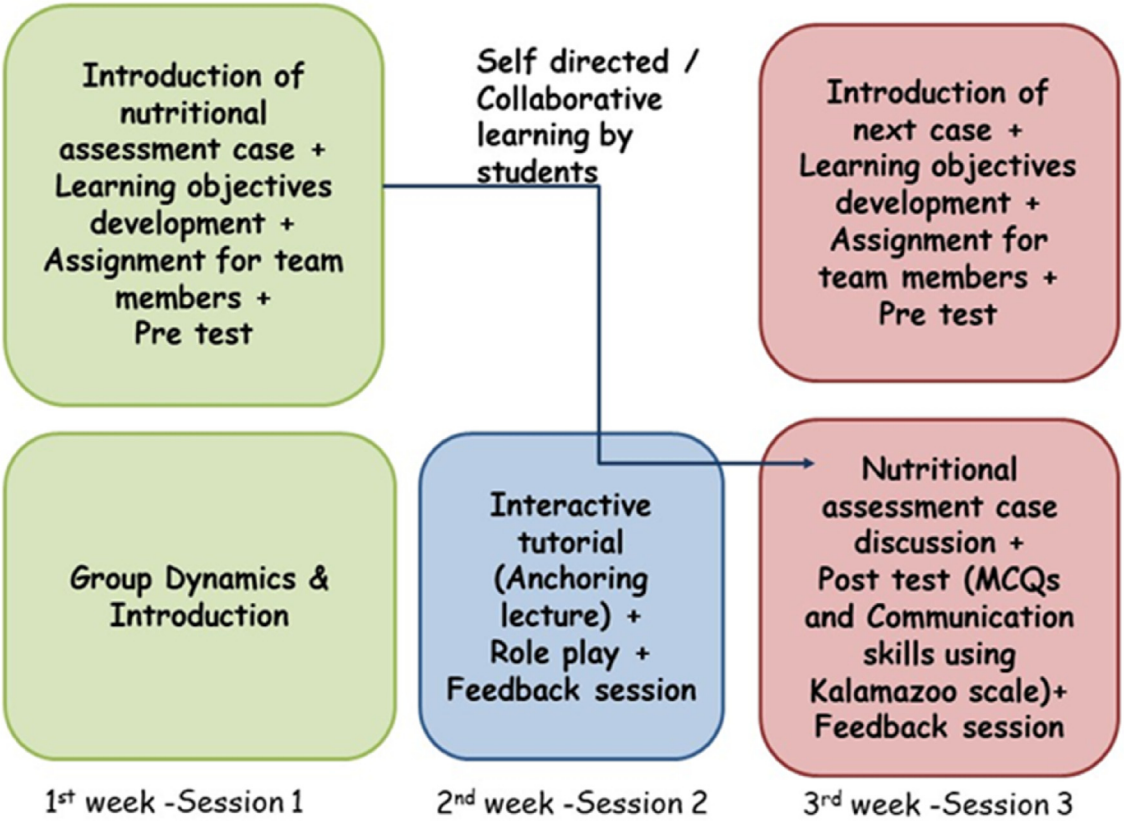


Fig. 1 – Hybrid problem-based learning model for nutritional assessment skills.

display of communication skills took place. In the 3rd week (session 3), case discussion (provided during session 1), post-test assessment, and feedback were conducted.

Feedback and suggestions were obtained from the study participants for quantitative (proforma) as well as qualitative (group discussions) assessments of training techniques. The evaluation was done on the basis of scores obtained by each participant on a test questionnaire and observation checklist.

The feedback of faculty was also obtained using a pre-structured questionnaire based on the Likert scale.

Evaluation

Data were coded, collected, and entered into MS–Excel software. Afterward, data were imported into SPSS version 21 software for analysis. Descriptive data were displayed as frequency and percentages. Non-parametric data comparison of means was done using the Mc–Nemar test (discrete) and Wilcoxon Signed Rank Test (continuous). A P value less than 0.05 is considered significant.

Informed consent of study participants was obtained. Personal Identification (anonymity) was not collected from the study participants. Privacy and confidentiality of data were maintained at all stages. The study was conducted after Institution Ethics Committee clearance (GMCH/GU/2019/1555).

Results

The study included 25 students of phase III of MBBS who were posted in the Department of Community Medicine for rotatory clinical posting. The institute is having an intake of 150 students every year and nearly 28 students are posted in each batch in various clinical departments including Community Medicine for approximately 4 weeks. The data of students who consented to study and attended all 3 sessions of HPBL were analyzed.

The mean age of study participants was 21.4 ± 1.9 years and comprised 9 (36%) males and 16 (64%) female participants. The Knowledge, practical, attitude, and communication aspects of nutritional assessment skills were recorded at baseline and after the HPBL session. On analysis, it was observed that there was a consistent and statistically significant improvement in knowledge, practice, attitude, and communication scores after HPBL (Table 1).

The satisfactory scores among students were assessed and we found a significant improvement in items across all three domains. However, a non-significant improvement was recorded in the knowledge of growth charting, severe acute malnutrition diagnosis, and practice of dietary assessment along with non-significant improvement in building up relationships, gathering and sharing information, reaching an agreement, and providing closure (Table 2).

Feedback from students was taken after the session and responses of the likert scale are shown in Fig. 2. The group discussion unveiled that the students had a very positive experience throughout. As the teaching program provided control to students in the learning process, it provided them with a sense of learning responsibility in similar lines to andragogical principles. Multiple aspects of the study sessions are the development of learning objectives, the team approach for collaborative learning, and debriefing with onsite case discussion and evaluation were comprehensive and inherently satisfactory for the students. Some of the student's comments highlight the learning advantages: "It was a totally new experience for me, never thought medical teaching can be so engaging. Another student mentioned "I was prepared to have a boring lecture and fill in a proforma for nutritional assessment and study later for exams, however, this approach is very student friendly we feel more responsible to achieve learning objectives which we have developed" to this other student agreed and reiterated the advantages of the hybrid model as it "allows students to understand the academic task or clinical problem in-depth and shared learning to reach optimal learning outputs". The students felt more enabled and competent in performing childhood nutritional assessment both in controlled and practice environments and that "similar methodology should be adopted in other subjects and topics i.e. child development assessment (paediatrics), management of common NCDs like hypertension and diabetes, etc (medicine).

Faculty feedback from faculty members has been represented in Fig. 3. When HPBL was first deployed, teachers were enthusiastic about it, but overall had some reservations about its application to a wide range of topics and capabilities. The method allowed students to have more control over their learning and improves their capacity to study more effectively while also increasing their motivation. On discussion, the teachers' responses revealed several advantages and a few limitations from a teaching and learning standpoint. A teacher mentioned that "Initially I was hesitant to emulate the technique as I was not sure if students will acquire the concepts and demonstrate deeper knowledge of the topic. However, the results are overwhelming and the experience is more than satisfactory from the faculty perspective". Other teachers highlighted significant advantages "Although the activity meant a greater time investment and follow-up, it served a better

Table 2 – Proportional satisfactory responses in pre- and post-intervention knowledge, practice, attitude and communication domains among students (n = 25).

Item (Scores)	Pre (%)	Post (%)	X^2, P value
1. Knowledge			
1. Growth chart (1)	15 (60)	21 (84)	2.08, 0.146
2. SAM diagnosis (1)	25 (100)	4 (16)	2.25, 0.134
3. Nutritional assessment methods (1)	14 (56)	22 (88)	4.90, 0.027
4. Dietary assessment methods (1)	11 (44)	19 (76)	4.78,0.022
5. Clinical sign/symptoms (5)	12 (48)	21 (84)	4.26, 0.039
2. Practice			
1. Medical history (5)	9 (36)	24 (96)	13.0, <0.001
2. Physical examination (5)	1 (4)	9 (36)	6.12, 0.013
3. Anthropometry (5)	7 (28)	24 (96)	15.05, <0.001
4. Dietary assessment (5)	6 (24)	13 (52)	2.76, 0.096
5. Clinical diagnosis and management (5)	0	8 (32)	6.12, 0.013
3. Attitude and communications			
1. Builds a relationship (5)	2 (8)	22 (88)	2.28, 0.131
2. Opens the discussion (5)	1 (4)	9 (36)	4.90, 0.027
3. Gathers information (5)	3 (12)	10 (40)	3.27, 0.070
4. Understands the patient's perspective (5)	0	7 (28)	5.14, 0.023
5. Shares information (5)	2 (8)	9 (36)	3.27, 0.070
6. Reaches agreement (5)	1 (4)	9 (36)	6.12, 0.013
7. Provides closure (5)	1 (4)	4 (16)	0.80, 0.375

Table 1 – Quantitative analysis of pre- and post-intervention knowledge, practice, attitude and communication scores of students (n = 25).

Item (Scores)	Pre (Mean ± SD)	Post (Mean ± SD)	Z, P value
1. Knowledge (9)	**4.76 ± 1.92**	**7.08 ± 1.03**	**3.81, <0.01**
1. Growth chart (1)	0.60 ± 0.50	0.84 ± 0.37	1.80, 0.083
2. SAM diagnosis (1)	0.00 ± 0.00	0.16 ± 0.37	2.13, 0.043
3. Nutritional assessment methods (1)	0.56 ± 0.50	0.88 ± 0.33	2.87, 0.008
4. Dietary assessment methods (1)	0.44 ± 0.50	0.76 ± 0.43	2.87, 0.008
5. Clinical sign/symptoms (5)	3.16 ± 1.37	4.44 ± 0.87	3.77, 0.001
2. Practice (25)	**9.92 ± 3.79**	**18.64 ± 3.01**	**4.37, < 0.01**
1. Medical history (5)	2.92 ± 1.25	4.36 ± 0.70	5.86, <0.001
2. Physical examination (5)	2.12 ± 1.16	3.48 ± 0.71	5.42, <0.001
3. Anthropometry (5)	2.68 ± 1.24	4.60 ± 0.57	7.10, <0.001
4. Dietary assessment (5)	2.12 ± 1.45	3.48 ± 1.19	4.23, <0.001
5. Clinical diagnosis and management (5)	0.08 ± 0.27	2.72 ± 1.64	7.74, <0.001
3. Attitude and communications(35)	**14.40 ± 5.52**	**21.56 ± 4.26**	**3.70, <0.01**
1. Builds a relationship (5)	2.36 ± 0.81	3.16 ± 0.62	3.84, 0.001
2. Opens the discussion (5)	2.16 ± 0.85	3.16 ± 0.94	3.78, 0.001
3. Gathers information (5)	2.08 ± 1.22	3.44 ± 0.71	5.03, <0.001
4. Understands the patient's perspective (5)	1.76 ± 1.10	3.16 ± 0.74	5.29, <0.001
5. Shares information (5)	2.12 ± 1.09	3.12 ± 1.16	3.16, 0.004
6. Reaches agreement (5)	2.04 ± 0.88	2.84 ± 1.06	2.88, 0.008
7. Provides closure (5)	1.88 ± 0.92	2.68 ± 0.90	3.36, 0.003

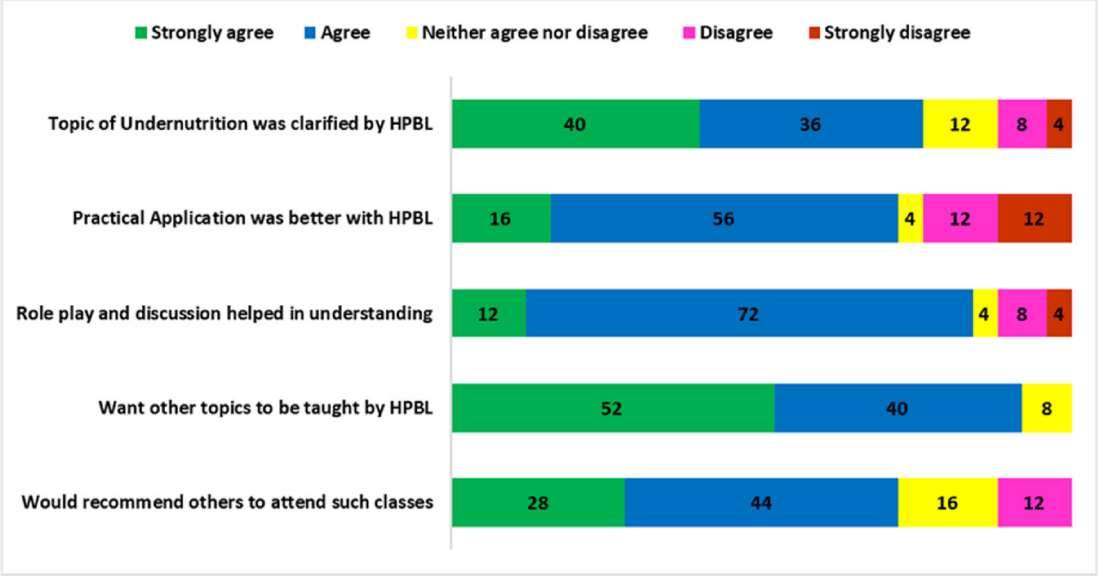


Fig. 2 – Feedback of students on the hybrid problem-based learning (HPBL) (n = 25).

contact, engagement, and rapport with the students which enhances the overall academic endeavor". Another added "It transforms the role responsibility structure of the process with teachers assuming a facilitator's role and students taking more responsibility for teaching. It is comforting as a teacher I used to feel responsible for teaching inputs and outputs". Though personally, most faculty were willing to adopt the methodology in their teaching practice they were skeptical if this will be suited to diverse subjects and topics as mentioned "This is a great one-time investment for curriculum planning HPBL into other community medicine topics as usually they involve both practical and theory aspects. However, it needs to be explored how will it be suitable to largely theoretical (pre-clinical), procedural and clinical competencies". Since the approach requires more resources (teacher's time, material, activity plans, tests, etc.) most faculty felt "it will be difficult to integrate fully in the curriculum yet, however, it is undeniably the stepping stone in the future of self-directed, participatory competency base medical education."

Discussion

The present study was planned to expand on the utility and prospects of HPBL as a student-centric medical education technique in practice-based teaching of medical students in field settings. We found that the approach not just improved factual knowledge acquisition but had significantly enhanced

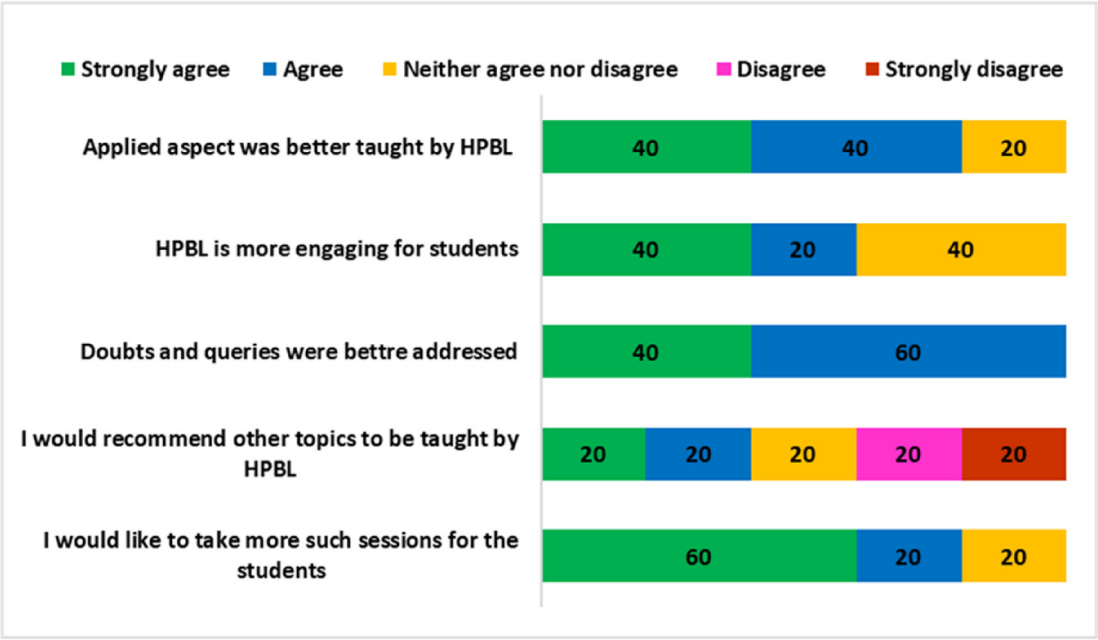


Fig. 3 – Feedback of faculty on the hybrid problem-based learning (HPBL) (n = 5).

practical attitude and communication skills among the learners. The approach was collaborative and participative, which ultimately drove the entire learning process in a flexible and contextual learning environment.[9] The novel teaching-learning method was assessed both in objective and experiential subjective measures and learning outcomes.

This study adopted a hybrid problem is learning technique to teach students nutritional assessment skills in community settings in a custom teaching program. The student engagement was above average with significant improvement recorded in knowledge, performance, attitude, and communication scores. Similar findings have been observed in previous studies where students felt HPBL was engaging and felt that it promoted collaboration and learning.[10,11] A similar research found that 2nd professional medical students actively learnt topics i.e. hypertension, diabetes, and so on with scope for self-directed learning during HPBL with enthusiastic faculty involvement.[12] In another experimental study where students who were oriented to problem-based learning PBL through brainstorming sessions had significantly better perception (cognitive, motivational scores) and performance (exam scores, success rate) than the control group who were not exposed to the PBL approach.[13]

Additionally, satisfactory skill improvements were observed in practice (medical history, anthropometry, etc.) and more specifically in attitude and communications aspects i.e. reaching agreement and understanding patients' perspectives, and so on. The HPBL approach had satisfactory performance across but could not achieve statistical significance in a few practical learning outcomes i.e. growth charting, diagnosis of SAM, and dietary assessment. Attitudes and communication aspects like building a relationship, gathering and sharing information, reaching an agreement, and providing closure were not significantly improved. This observation may be attributed to the short-term follow-up of HPBL on learning outcomes as stated by previous literature on the effectiveness of PBL. Furthermore, most research implies that the PBL does not always have a direct influence on immediate skills gathering, but rather emphasizes problem-solving skills and increases the skills of applying knowledge in a long-term.[14]

Feedback from students and faculty was collected and discussions were held to understand the perceptions and subjective experiences with the HPBL program. The majority of students provided positive feedback and recommended the inclusion of HPBL in regular teaching sessions during their clinical postings. Students generally agreed that HPBL is a more pragmatic teaching-learning method in terms of clarity, practical application, and discussion for acquiring definitive skills such as nutritional assessment, and so on. The students unequivocally appreciated better control provided to them in the whole learning process giving them a sense of responsibility, self-direction, and flexibility. Similarly, most faculty members agreed that applied aspects were better taught with higher student engagement in HPBL and were willing to take more of such sessions. The faculty members were a bit skeptical considering the greater time investment in planning a new course structure and the requirement of a follow-up in place of traditional single contact teaching. The finding of our study elaborates on the similar observations reported in previous studies.[6,13] As reported in a previous study majority (65%) of faculty members reported a preference for PBL, which was significantly higher among clinical faculty than those from pre and para clinical specialties.[15]

Implications of the study: The current study can provide directions for planning HPBLs in other subjects and disciplines by faculty development programs. After accumulating sufficient evidence, HPBL methods may be suggested in future medical curricula.

Our study has several limitations. The most important is the absence of a control/standard teaching method to compare the teaching methods and estimate additional benefits attained by learners by the HPBL approach. The sample size of the present educational research was small considering its exploratory and interventional nature. The study incorporated qualitative methods for broader insights into the study's pivotal outcomes paving way for similar research projects with larger sample sizes.

The long-term assessment of learning outcomes was not possible within the scope of study which could have reflected on knowledge and skill retention among learners. Also, since there is a lack of evidence in the literature about the effect of an HPBL on students' learning outcomes, we did not assess possible confounding factors that could have possibly affected the study results.

Conclusion

Our study demonstrates that HPBL is a much-suited teaching methodology especially for imparting clinical skills such as nutritional assessment in simulated environments. This offers a unique blend of traditional instruction-based learning to flexible collaborative learning which is more outcome-based. The present study unveiled the learning advantage offered by HPBL in enhancing knowledge, practical and especially attitudinal and communication skills as observed in the study participants. Subjective assessment in the study suggested students were receptive to the new learning method which ultimately resulted in better learning outcomes. Whereas, faculty members also suggested that the HPBL method can be very instrumental in achieving desired learning objectives, especially in contextual settings and when supported with resources and a faculty development program.

Disclosure of competing interest

The authors have none to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.mjafi.2022.06.017.

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